# Course Manager • Instructor • Learner

## Three Dashboard Relationship Guide

How the Three Core Dashboards Connect, Share Data, and Work Together

PhysioGuidance • NELC-Compliant • Bilingual AR/EN • WCAG 2.1 AA

WORKING ASSUMPTION: Course Manager creates courses and assigns instructors. Instructor adds content and delivers. Learner consumes. This flow is pending client (Amna) approval and may change to instructor-creates model.

| Role | Dashboard Purpose | Data Scope |
|---|---|---|
| **Course Manager** | Plan courses, assign instructors, review quality, ensure NELC compliance | ALL courses, ALL instructors, platform-wide |
| **Instructor** | Add content, deliver sessions, track students, grade assessments | OWN assigned courses and enrolled students only |
| **Learner** | Enroll, study, attend sessions, take assessments, earn certificates | OWN enrollments, progress, and certificates only |

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# 1. The Big Picture — How Three Dashboards Connect

These three dashboards are not independent. They share the same database tables, the same courses, the same sessions, and the same students. What changes is the perspective and the permissions.

## 1.1 The Complete Course Lifecycle

This is the master flow that connects all three roles. Every action in one dashboard creates data that appears in another:

1. **Course Manager creates a course** (title, description, schedule, delivery type, NELC requirements, max enrollment)
2. **Course Manager assigns an Instructor** to the course (sets instructorId on the course)
3. **Instructor receives the course** in their dashboard and adds content: uploads videos/PDFs, creates modules, builds assessments, writes questions
4. **Course Manager reviews the content** (NELC checklist, WCAG accessibility, objective alignment). Approves or rejects with feedback.
5. **Course goes ACTIVE** — now visible to learners in the course catalog
6. **Learner browses and enrolls** — enrollment record created, appears in both instructor and CM views
7. **Instructor schedules sessions** (Zoom/on-site) — appears in learner's calendar
8. **Learner attends sessions** — instructor marks attendance — attendance data flows to CM reports
9. **Learner takes assessments** — auto-graded MCQ or instructor-graded essays — scores visible to all three
10. **Learner completes course** (progress = 100%) — certificate request triggers
11. **Instructor reviews certificate eligibility** (attendance ≥70%, all assessments passed) — approves/rejects
12. **Learner downloads certificate** with QR code — CM sees issuance in platform reports

## 1.2 Data Flow Diagram

This shows which role WRITES data and which role READS it:

| Data | Who CREATES It | Who READS It | Database Table |
|---|---|---|---|
| **Course** | Course Manager | CM + Instructor + Learner | courses |
| **Instructor Assignment** | Course Manager | CM + Instructor | courses.instructorId |
| **Modules** | Instructor | Instructor + Learner | course_modules |
| **Materials** | Instructor (uploads) | CM (reviews) + Learner (views) | course_materials |
| **Sessions** | Instructor (schedules) | CM (oversight) + Learner (attends) | sessions |
| **Enrollment** | Learner (enrolls) | CM + Instructor + Learner | enrollments |
| **Attendance** | Instructor (marks) | CM (reports) + Learner (views own) | attendance |
| **Assessments** | Instructor (creates) | CM (oversight) + Learner (takes) | assessments |
| **Questions** | Instructor | Learner (answers) | questions |
| **Submissions** | Learner (submits) | Instructor (grades) + Learner (views) | submissions |
| **Certificate Request** | System (auto-triggers) | Instructor (reviews) + Learner (downloads) | certificate_requests |
| **NELC Compliance** | System (auto-checks) | Course Manager | computed / audit logs |
| **Reports** | System (generates) | CM (platform-wide) + Instructor (own courses) | computed exports |

## 1.3 Shared Database Tables

All three dashboards read from the SAME tables. The difference is the WHERE clause:

| Table | Course Manager Sees | Instructor Sees | Learner Sees |
|---|---|---|---|
| **courses** | ALL courses (any status) | Only WHERE instructorId = me | Only WHERE status = 'active' (catalog) |
| **sessions** | ALL sessions (all instructors) | Only WHERE instructorId = me | Only WHERE enrolled in that course |
| **enrollments** | ALL enrollments (platform-wide) | Only for my assigned courses | Only WHERE studentId = me |
| **attendance** | ALL records (for reports) | Only for my sessions | Only WHERE studentId = me |
| **submissions** | Overview stats only | For my course assessments | Only WHERE studentId = me |
| **certificate_requests** | Stats (issued count, rate) | For my assigned courses | Only WHERE studentId = me |
| **course_materials** | ALL (for review/approval) | Only for my courses | Only for enrolled courses (approved only) |

# 2. Feature-by-Feature Comparison

This section maps every feature and shows which role does what. This is the key reference for your developers to understand scope boundaries.

## 2.1 Courses

| Action | Course Manager | Instructor | Learner |
|---|---|---|---|
| **Create course** | YES — creates structure, schedule, NELC config | NO | NO |
| **Assign instructor** | YES — sets instructorId | NO (receives assignment) | NO |
| **Edit course info** | YES — title, dates, max enrollment | Limited — can suggest changes | NO |
| **Add modules** | NO (defines structure only) | YES — adds content to modules | NO |
| **Upload materials** | NO | YES — uploads video/PDF/docs | NO |
| **Review content** | YES — approve/reject before publish | NO (submits for review) | NO |
| **View course catalog** | YES (all courses) | YES (own courses) | YES (active courses to browse & enroll) |
| **Enroll in course** | NO | NO | YES — the ONLY role that enrolls |
| **Delete course** | YES (drafts only) | NO | NO |

## 2.2 Sessions

| Action | Course Manager | Instructor | Learner |
|---|---|---|---|
| **Schedule session** | Can set initial schedule during course creation | YES — schedules all sessions | NO |
| **Start session** | NO | YES — one-click Start | NO |
| **Join session** | NO (oversight only) | YES (as host) | YES — joins Zoom or attends on-site |
| **Mark attendance** | NO | YES — marks present/absent/late | NO |
| **View attendance** | YES (all sessions, all instructors) | YES (own sessions) | YES (own attendance only) |
| **Cancel** | YES (can cancel on | YES (own | NO |

| | | | |
|---|---|---|---|
| **session** | behalf of instructor) | sessions) | |
| **View calendar** | YES (all instructors' schedules) | YES (own sessions) | YES (enrolled sessions only) |

## 2.3 Assessments & Grading

| Action | Course Manager | Instructor | Learner |
|---|---|---|---|
| **Create assessment** | NO | YES — creates quizzes, exams, assignments | NO |
| **Create questions** | NO | YES — MCQ, T/F, essay, short answer | NO |
| **Set retake rules** | YES — defines policies per course | Follows CM's rules | NO |
| **Take assessment** | NO | NO | YES — the ONLY role that takes assessments |
| **Submit answers** | NO | NO | YES |
| **Auto-grade MCQ/TF** | N/A | System does this automatically | Sees score immediately |
| **Manual grade essay** | NO | YES — grades essays/short answers | Sees score after grading |
| **View item analysis** | YES (oversight) | YES (own assessments) | NO |
| **View own scores** | NO | NO | YES — sees scores in transcript |

## 2.4 Certificates

| Action | Course Manager | Instructor | Learner |
|---|---|---|---|
| **Request certificate** | NO | NO | Auto-triggered when course complete |
| **Review eligibility** | Dashboard stats only | YES — reviews attendance %, scores | Views own eligibility status |
| **Approve/ reject** | NO | YES — approves or rejects with note | NO |
| **Download certificate** | NO | NO | YES — downloads PDF with QR code |
| **Verify certificate** | Public QR page | Public QR page | Public QR page (shares link) |
| **View issuance stats** | YES (platform-wide) | YES (own courses) | NO |

## 2.5 Reports & Compliance

| Action | Course Manager | Instructor | Learner |
|---|---|---|---|
| **Attendance report** | YES — all courses, NELC evidence | YES — own courses only | NO (sees own attendance %) |
| **Gradebook export** | YES — platform-wide | YES — own courses | NO (sees own transcript) |
| **NELC compliance** | YES — full compliance center | NO | NO |
| **Enrollment report** | YES — all courses | YES — own courses | NO |
| **Export CSV/PDF** | YES (all reports) | YES (own reports) | Limited (own transcript) |

# 3. Learner Dashboard — Complete Feature Breakdown

Role: STUDENT | Authorization: Own data only (WHERE studentId = currentUser.id) | Purpose: Simple, motivating view of plan, progress, and achievements

## 3.1 Learner KPI Cards

| Card | What It Shows | Data Source |
|---|---|---|
| **Overall Progress** | % completion across all active courses | AVG(enrollments.progress) WHERE studentId = me |
| **Attendance Rate** | Current attendance vs required (70% NELC min) | count(present) / count(sessions) for enrolled courses |
| **Certificates Earned** | Issued + pending eligibility + expiring soon | certificate_requests WHERE studentId = me |
| **Upcoming Sessions** | Next session date and join button | sessions WHERE enrolled AND scheduledAt > now |

## 3.2 My Courses (Learner View)

Cards showing all enrolled courses with clear next-action prompts:

- Resume lesson — shows last module/video where learner left off
- Join Zoom — if a live session is happening or starting soon (within 15 min)
- Take quiz — if an assessment is published and not yet attempted
- Upload assignment — if an assignment assessment is pending submission
- Visual progress bar (0-100%) per course
- Deadline alerts: red badge for overdue tasks, yellow for upcoming
- Course status: In Progress / Completed / Dropped

## 3.3 Calendar & Reminders

- Monthly calendar showing upcoming sessions with join buttons
- Timezone-aware: session times displayed in learner's local timezone
- AR/EN toggle for calendar interface
- Reminders via email/SMS/WhatsApp (using Unifonic/Infobip integration)
- Snooze or mark done on each reminder
- Reminder timing: 1 hour, 30 min, 15 min before session (configurable)

## 3.4 Transcripts & Certificates

### 3.4.1 Academic Transcript

- Lists all completed and active courses

- Per course: assessment scores, attendance percentage, overall grade
- Cumulative GPA/average if applicable
- Downloadable as PDF

### 3.4.2 Certificates

- Certificate status: Eligible (pending review), Approved (download ready), Not yet eligible
- Download certificate as PDF
- Each certificate has: unique certificate ID, QR code, digital signature
- Public verification page: anyone can scan QR or enter certificate ID to verify authenticity
- Expiring soon alerts (if certificates have validity periods)

## 3.5 Support & Help

- Open a support ticket with priority (High/Medium/Low)
- Ticket categories: Billing, Login, Content, Certificate, Technical
- View ticket status and SLA countdown (connects to Support Dashboard)
- FAQ section with searchable common questions
- Quick video tutorials for common tasks (how to enroll, how to join Zoom, etc.)

## 3.6 Accessibility & Preferences

- Language: Arabic / English toggle
- Font size: Small / Medium / Large
- High contrast mode toggle
- Captions/subtitles: on by default for all video content
- Privacy controls: data sharing opt-in/out, notification preferences
- Notification channels: choose email, SMS, WhatsApp, or all

# 4. Separate vs Shared Features

This is the most important section for your developers. It tells them what can be built as shared components and what must be role-specific.

## 4.1 Shared Backend (Same Tables, Same APIs)

These backend modules serve ALL three dashboards. Build ONCE, filter by role:

| Backend Module | Tables | Used By | Filter Logic |
|---|---|---|---|
| **Courses Module** | courses, course_modules | CM (all) + Instructor (own) + Learner (active catalog) | Role-based WHERE clause |
| **Sessions Module** | sessions | CM (all) + Instructor (own) + Learner (enrolled) | Role-based WHERE clause |
| **Enrollments Module** | enrollments | CM (all) + Instructor (own courses) + Learner (own) | Role-based WHERE clause |
| **Attendance Module** | attendance | CM (all) + Instructor (own sessions) + Learner (own) | Role-based WHERE clause |
| **Assessments Module** | assessments, questions, submissions | Instructor (CRUD) + Learner (take/submit) | Role determines read/write |
| **Materials Module** | course_materials | Instructor (upload) + CM (review) + Learner (view approved) | Status filter for learner |
| **Certificates Module** | certificate_requests | Instructor (review) + Learner (request/download) | Role determines actions |
| **Reports Module** | computed from above | CM (platform) + Instructor (own) | Scope parameter |

## 4.2 Role-Specific Features (Build Separately)

| Feature | Role | Why It's Separate |
|---|---|---|
| **Course creation form** | Course Manager ONLY | Only CM creates courses in this flow |
| **Instructor assignment** | Course Manager ONLY | Only CM assigns instructors to courses |
| **Content review queue** | Course Manager ONLY | Approval workflow unique to CM |
| **NELC Compliance Center** | Course Manager ONLY | Compliance monitoring is CM responsibility |
| **Instructor roster & KPIs** | Course Manager ONLY | Only CM monitors instructor performance |

| | | |
|---|---|---|
| **Content upload/versioning** | Instructor ONLY | Only instructors upload course materials |
| **Session start/end** | Instructor ONLY | Only instructor controls live sessions |
| **Grading interface** | Instructor ONLY | Only instructor grades essays/assignments |
| **At-risk learner interventions** | Instructor ONLY | Instructor sends reminders, schedules make-ups |
| **Course enrollment** | Learner ONLY | Only learners enroll themselves |
| **Assessment taking UI** | Learner ONLY | Quiz/exam interface unique to learner |
| **Certificate download + QR** | Learner ONLY | Only learner downloads their own certificate |
| **Support ticket creation** | Learner ONLY | Only learner opens support tickets |
| **Progress tracking UI** | Learner ONLY | Personal progress bars, next-action prompts |

## 4.3 Shared Frontend Components (Reusable)

These UI components can be built ONCE and reused across dashboards with different data:

| Component | Used By | Differences Per Role |
|---|---|---|
| **KPI Cards** | All three dashboards | Different metrics, same card design |
| **Calendar Widget** | All three dashboards | CM sees all, Instructor sees own, Learner sees enrolled |
| **Course Card** | All three dashboards | CM has assign/review actions, Instructor has edit, Learner has enroll/resume |
| **Session Card** | All three dashboards | CM has oversight, Instructor has start/end, Learner has join |
| **Data Table** | All three dashboards | Same table component, different columns and filters |
| **Export Button** | CM + Instructor | Same CSV/PDF export logic, different data scope |
| **Status Badges** | All three dashboards | Same badge component (Active/Draft/Completed) |
| **Search + Filters** | All three dashboards | Same filter UI, different filter options per role |
| **Sidebar Navigation** | All three dashboards | Same shell, different menu items per role |
| **Profile Card** | All three dashboards | Same layout, different role badge |

# 5. Recommended Build Order for Developers

Since the three dashboards share backend tables, the build order matters. You cannot build the Instructor dashboard without the courses table, and you cannot build the Learner dashboard without enrollment and session tables.

## 5.1 Phase 1 — Shared Backend (Build First)

These modules serve all three dashboards. Build them ONCE with role-based filtering:

13. **Courses + Modules schema + CRUD** (courses, course_modules tables — used by all 3)
14. **Sessions schema + CRUD** (sessions table — used by all 3)
15. **Enrollments + Attendance schema + CRUD** (enrollments, attendance tables — used by all 3)
16. **Assessments + Questions + Submissions** (3 tables — Instructor creates, Learner takes)
17. **Materials + File Upload** (course_materials table — Instructor uploads, CM reviews, Learner views)
18. **Certificates** (certificate_requests table — Instructor reviews, Learner downloads)
19. **Reports engine** (shared export logic — CM and Instructor use with different scopes)

## 5.2 Phase 2 — Course Manager Dashboard (Build Second)

CM dashboard is built second because it creates courses that instructors and learners interact with:

20. **CM Dashboard Home** (KPIs, catalog overview, content review queue)
21. **Course Creation Form** (multi-step: info, schedule, NELC config, instructor assignment)
22. **Instructor Roster + Assignment** (list instructors, assign to courses, monitor KPIs)
23. **Content Review Workflow** (approve/reject materials before publish)
24. **NELC Compliance Center** (compliance scorecard, evidence export)
25. **CM Reports** (platform-wide analytics, NELC evidence bundles)
26. **Schedules Oversight** (cross-instructor calendar, conflict detection)

## 5.3 Phase 3 — Instructor Dashboard (Build Third)

Instructor dashboard depends on courses existing (created by CM) and shared backend tables:

27. **Instructor Dashboard Home** (KPIs, today's sessions, at-risk learners, cert queue)
28. **My Courses + Course Detail** (6 tabs — view assigned courses, add content)
29. **Content Upload + Versioning** (upload materials, submit for CM review)
30. **Sessions Management** (schedule, start/end, attendance tracking)
31. **Students View** (enrolled students, at-risk detection, interventions)

32. **Assessments + Question Bank** (create assessments, grade submissions, item analysis)
33. **Certificate Review** (approve/reject/hold, batch operations)
34. **Reports** (attendance, gradebook, enrollment, completion — own courses)

## 5.4 Phase 4 — Learner Dashboard (Build Last)

Learner dashboard is the consumer of everything created by CM and Instructor:

35. **Learner Dashboard Home** (KPIs, my courses cards, upcoming sessions)
36. **Course Catalog + Enrollment** (browse active courses, enroll)
37. **My Courses View** (progress bars, next-action prompts, deadlines)
38. **Session Calendar + Join** (calendar, Zoom join button, reminders)
39. **Assessment Taking UI** (quiz interface, timer, auto-submit, score display)
40. **Transcript + Certificates** (academic record, certificate download, QR verification)
41. **Support & Help** (ticket creation, FAQ, tutorial videos)
42. **Accessibility Preferences** (language, font, contrast, captions, privacy)

# 6. What Changes If Client Approves Option A

IMPORTANT: This entire document assumes Option B (CM creates courses, assigns instructors). If client Amna approves Option A (Instructor creates courses, CM reviews), the following changes apply:

| Feature | Option B (Current Doc) | Option A (If Approved) |
|---|---|---|
| **Course creation** | Course Manager creates | Instructor creates |
| **Instructor assignment** | CM assigns instructor to course | N/A — instructor owns their own course |
| **Course ownership** | courses.createdBy = CM, courses.instructorId = assigned | courses.instructorId = creator (same person) |
| **Course status flow** | Draft → Active (CM publishes) | Draft → Pending Review → Active (CM approves) |
| **CM 'Create Course' page** | Full course creation form | REMOVED — CM has no creation form |
| **CM 'Assign Instructor' page** | Instructor roster with assign buttons | REMOVED — becomes read-only monitoring |
| **Instructor 'My Courses'** | Shows assigned courses (read-only structure) | Shows own courses (full edit access) |
| **Content review** | Same — CM reviews before publish | Same — CM reviews before publish |
| **Everything else** | Same | Same — sessions, enrollment, attendance, assessments, certificates unchanged |

The backend tables remain the same in both models. The only changes are: (1) which role has POST /courses permission, (2) whether the instructor assignment endpoint exists, and (3) whether courses need a 'pending_review' status. This is a frontend + permissions change, not a database redesign.

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